

Certified Material Test Report

Cert #: 302055 Mill Order: 1824430 Heat #: A185465 Issued: 1/2/2019 00:18:10
Work Order: 272327 Sales Order: 208477-1 Customer: American Alloy Steel PO #: 116892-1
Load #: 320940 Reference #: Reference Desc: End Use:
Size: 1-3/4" Shape: Round Grade: A350/LF2 Chem Length: 28'
Grain Practice: Al Fine Grain (5-8) per ASTM A29 Reduction Ratio: 60.4 to 1 Disposition: Rolled Prime

Ladle Chemistry Analysis (ASTM A29)

C	Mn	P	S	Si	Al	Cu	Ni	Cr	Mo	Sn	N	V	Cb	B	Ca	W	Ti	DI
0.18	1.14	0.011	0.004	0.24	0.030	0.20	0.08	0.11	0.03	0.008	0.0063	0.031	0.000	0.0000	0.0010	0.000	0.001	0.86
Pb	Co	As	Sb	Zr	Bi	H (ppm)	O (ppm)	Ceq	J-Factor									
0.000	0.006	0.004	0.003	0.000	0.000	1.4		0.42	262									

Product Check Analysis (ASTM A29)

C	Mn	P	S	Si	Al	Cu	Ni	Cr	Mo	Sn	N	V	Cb	Ti	B	Ca	O
Front																	
Back																	

Jominy (ASTM A255)

	J1	J2	J3	J4	J5	J6	J7	J8	J9	J10	J12	J14	J16	J18	J20	J24	J28	J32
Calc'd Standard	1.5	3	5	7	9	11	13	15	20	25	30	35	40	45	50			
Calc'd Metric																		
	J1	J2	J3	J4	J5	J6	J7	J8	J9	J10	J12	J14	J16	J18	J20	J24	J28	J32
Front																		
Back																		

Microcleanliness (ASTM E45)

Method A								Method C (SAE J422)		Method E	
AT	AH	BT	BH	CT	CH	DT	DH	S	O	SAM "B"	SAM "D"

Microcleanliness (DIN 50602)

K			M	
S	O	Tot	Tot	

Decarb		Grainsize		Macrostructure (ASTM E381)			Magnetic Particle Inspection	
Depth	% of Diameter	Austenitic	Ferritic	S	R	C	Frequency	Severity

Mechanical Properties (ASTM A370)

Tensile Properties					Hardness	
Tensile Strength	0.2% Yield Strength	% Elong (2")	% ROA	0.35% EUL Yield Strength	(MR)	(Surf)

Steel Dynamics - Engineered Bar Products has a quality system in place which has been certified ISO 9001:2015 compliant, including PED certification.

Comments/Specs

Electric Arc Furnace Melted - Vacuum Tank Degassed --- Normalize and Machine Straighten --- Normalize and Machine Straighten --- ASTM A350-12 LF2 Class 1 --- ASTM A696-90a Grade C --- ASTM A675-03 Grade 70 --- ASME SA350-04 LF-2 Class 1 --- ASME SA105-04 --- ASME SA696-04 Gr C --- ASME SA675-04 Gr 70 --- Complies with EN 10204 type 3.1 --- Pres Equip Direct(PED) 97/23/EC/7/2 Annex I, Par 4.3 --- ASTM A105/A105M -18

Certified true copy of the
original retained in our file.
AMERICAN ALLOY STEEL

Condition: Normalize, Straighten

I hereby certify that the content of this report is correct and accurate, and that all tests and operations performed on this material were in compliance with applicable material specifications and purchaser designated requirements.

Jonathan Vallosio - Rolling Mill Metallurgist (ES)

Any alteration to this report voids Steel Dynamic's warranting of results. No weld repair has been performed on this material. This material is not radioactive and has not been exposed to radioactivity while under the control of Steel Dynamics. This material has not been exposed to mercury while under the control of Steel Dynamics. Unless otherwise noted, this material was melted, continually cast, and rolled in the USA; w/ all testing performed by Steel Dynamics.

1-3/4" Dia. SA-350 LF-2/SA-105/SA675-70/SA-696-C
Dark Green
Silver
HT. A185465